Andrew Marion

Marietta, GA | 678-615-9817 | drew.marion03@gmail.com | Portfolio | LinkedIn | ResearchGate

EDUCATION

KENNESAW STATE UNIVERSITY (KSU) | Marietta, GA — Anticipated Graduation May 2027

• *Bachelor of Science in Mechanical Engineering, Aerospace Engineering Minor* — GPA: **3.81/4.0**

RELEVANT EXPERIENCE

NASA Glenn Research Center | Cleveland, OH — August 2026 – December 2026

UAV Flight Control Intern, Diagnostics and Controls Branch

- Implement custom MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 autopilot for a 12 ft wingspan, 16-fan distributed electric propulsion (DEP) UAV based on NASA's SUSAN Electrofan configuration
- Develop fault-tolerant control allocation algorithms in MATLAB that redistribute thrust after fan or thrust-vectoring failures, evaluated across 8,000 Monte Carlo fan-failure cases
- Model UAV flight dynamics and powertrain behavior in MATLAB/Simulink and X-Plane for control law development

Textron GSE | Cartersville, GA — June 2026 – August 2026

Mechanical Design Engineer Intern, Ground Support Equipment Electrification

- Designed 3 assemblies and 15 parts in Siemens NX with production drawings for electric ground support equipment, including the MX4 baggage tractor hinged door and 660 belt loader access doors
- Ran motion simulation and prototype tests, including parking brake sensor placement and door kinematics, verifying function and reducing design rework before production release

RESEARCH EXPERIENCE

KSU Advanced Aerodynamics and Propulsion Lab | Marietta, GA — September 2023 – Present

Undergraduate Research Assistant

Numerical Investigation of Compound Delta Wings — September 2023 – Present

- Executed 140+ RANS CFD simulations in ANSYS Fluent across Mach numbers and angles of attack, increasing lift-to-drag ratio 7.24–13.57% over the baseline wing in the transonic regime

Control Surface Validation of the "Bird" UAV with FlightStream — March 2025 – Present

- Evaluated mid-fidelity control surface modeling in FlightStream with Oklahoma State's S2FAR Lab and Siemens
- Validated FlightStream against CFD, flight test, and WindShape wind tunnel data; published at AIAA SciTech 2026

Aerodynamic Characterization of a Blended Wing Body Using FlightStream — January 2026 – Present

- Benchmarked FlightStream against a 42-case transonic RANS sweep (Mach 0.6–0.8); FlightStream ran 5× faster with lift coefficient within 0.013 in the linear range

PUBLICATIONS

- Marion, A., Sharma, G., et al., "Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027
- Atkinson, Z., Comer, A., Marion, A., et al., "Multi-Fidelity Validation of Control Surface Deflection Predictions Using FlightStream," AIAA SciTech 2026. doi.org/10.2514/6.2026-2810
- Plus 4 more papers accepted to AIAA SciTech 2027 and a first-author 2025 AIAA Region II Student Conference paper (doi.org/10.2514/6.2025-97857)

LEADERSHIP AND COMPETITION TEAMS

Kennesaw Rocketry Team | Marietta, GA — May 2025 – June 2026

Vice President (December 2025 – June 2026); Liquid Propulsion Lead (May 2025 – December 2025)

- Led team operations as VP; led initial design and sizing of a 250 lbf N2O/ethanol engine (O/F 4.8) as Liquid Lead

Aerial Robotics Competition Team | Marietta, GA — September 2023 – April 2025

PADA Team Lead

- Led 20+ members through design, manufacturing, and flight testing of a powered autonomous delivery aircraft (PADA), placing 10th of 19 teams in the Advanced Class at SAE Aero Design 2024
- Configured and tuned a Matek H743 flight controller running ArduPilot for autonomous flight of an inherently unstable flying wing that released from a carrier aircraft mid-flight and navigated to target landing coordinates

TECHNICAL SKILLS

CFD and Aerodynamics: ANSYS Fluent, OpenFOAM, FlightStream, SolidWorks Flow Simulation, OpenVSP, ParaView
Controls and Programming: MATLAB/Simulink, Python, R, LaTeX, PX4, ArduPilot, Pixhawk, X-Plane
CAD: Siemens NX, SolidWorks, AutoCAD, ANSYS SpaceClaim, GD&T, Tolerance Stack-Up, DFA/DFM
Manufacturing: 3D Printing, Laser Cutting, Composite Layup, Lathe/Mill, CNC Machining, Foam Cutting